

Saransh Gupta

Experienced Data Scientist with 4+ years of expertise in delivering AI/ML-driven solutions across enterprise environments. Proven track record as a researcher, with a strong focus on applying machine learning to real-world problems, driving innovation, and contributing to strategic initiatives that align with organizational goals.

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in [LinkedIn Profile](#)



Executive Profile

- Dedicated professional offering more than 4 years of experience in developing machine learning models and transforming data science prototypes into production grade solutions.
- Assessed strategies and validate modifications in machine learning models to enhance NLP systems continually, ensuring consistent improvement.
- Understanding of the concept of data science - advanced analytics, predictive modeler, machine learning algorithm in multiple technical and functional domains.
- Expertise in conducting full lifecycle analysis including data gathering and cleansing, deep dive advanced statistical analysis/modeling and recommendations to optimize performance.
- Skilled in leveraging Python, PyTorch, Transformers, BERT, scikit-learn, and TensorFlow to develop advanced machine learning models and optimize NLP systems.
- A focused individual with a zeal to learn and adapt to new technologies quickly; capabilities in managing critical situation.



Education & Certifications

- Indian Institute of Technology Kharagpur (2017 – 2022)
B. Tech + M. Tech in Engineering Product Design, Industrial and Systems Engineering
Grade: 8.09 / 10
- AWS Certified Cloud Practitioner ([verify](#))



Publications

- "A hybrid approach for Anomaly prediction in server performance using clustering, outlier detection and gradient boosting", DevCon 2025, American Express
- "ClotCatcher: A Novel Natural Language Model to Accurately Adjudicate Venous Thromboembolism from Radiology Reports" BMC Medical Informatics and Decision Making doi: [10.1186/s12911-023-02369-z](https://doi.org/10.1186/s12911-023-02369-z)
- S. Gupta et al. "Integrative Network Modeling Highlights the Crucial Roles of Rho-GDI Signaling Pathway in the Progression of Non-Small Cell Lung Cancer" in IEEE - JBHI, 2022, doi: [10.1109/JBHI.2022.3190038](https://doi.org/10.1109/JBHI.2022.3190038)
- Entity-aware Question-Answer Extraction for Shopping Guidance, Amazon Machine Learning Conference



Achievements

- Conferred with an exceptional performance rating at the American Express for impactful contribution to the organization in year 2024
- Received scholarship of 248 USD for Harvard College Project for Asian International Relations conference - 2022
- Featured as one of the Top 30 Undergraduate Achievers of IIT Kharagpur in the UG Achievers Directory 2020
- Awarded scholarship of 2200€ by The A*Midex Foundation of Aix-Marseille University, France, Feb 2020
- Selected among Top 5 percent out of all for the summer fellowship at Institute of Science Technology Austria
- Featured in the ISE Newsletter Autumn-2020 under Department Spotlight of ISE fights COVID- 19, 2020



Links

- [GitHub](#)
- [Google Scholar](#)
- [Website](#)

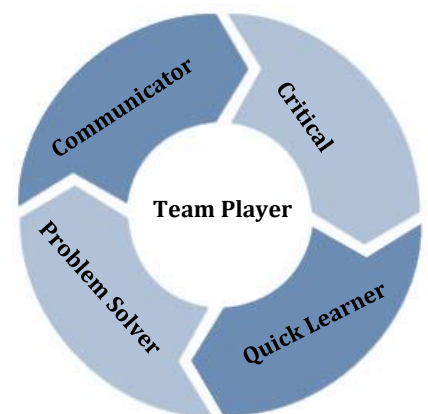


Core Competencies

Natural Language Processing	Machine Learning Algorithms
Data Visualization	Agentic AI
Predictive Modeling	Amazon Web Services
SQL	Text Mining
Deep Learning	Neural Networks



Soft Skills



Technical Skills

- Python
- PyTorch
- Transformers
- BERT
- Large Language Models
- Transfer Learning
- scikit-learn
- TensorFlow
- Amazon Web Services
- SQL
- GCP
- AWS



Professional Experience

American Express | AI Engineer-II | April 2024 – Present

Project: ServiceNow ITSM Copilot

- Engineered a RAG system using GPT-4.1 suggesting the incident resolution steps and reducing manual work-around time by over 60%
- Developed an intelligent incident-copilot pipeline that retrieves top 10 semantically similar historical tickets from a vector database
- Leveraged semantic embedding and vector search to provide LLM-based suggestions, leading to an overall 20% improvement in the SLA
- Deployed the RAG system in production servers of the American Express ServiceNow ITSM

Project: Identify Anomalous activities on company servers

- Designed and implemented a robust anomaly detection system to identify unusual patterns in server logs
- Applied advanced unsupervised learning algorithms such as Local Outlier Factor (LOF), Isolation Forest, and DBSCAN to detect anomalies
- Achieved F1 score of 0.91 in identifying anomalies by further training with XGBoost model on the labeled anomalies
- Demonstrated capabilities of this model along with server statistics in a UI built in streamlit

American Express | Engineer-III | Aug 2022 – April 2024

Project: Failure cause identification of applications on generated Incident for their automated resolve

- Implemented a Question-Answer based strategy on top of raw dataset to identify failure cause of applications
- Achieved F1 Score of 0.84 by fine tuning a pre-trained BERT based Question-Answering model

Tools and Software: Python, PyTorch, GPT-4.1, BERT, Pandas, NumPy, matplotlib, SQL, streamlit, ServiceNow

Amazon Development Centre India | Applied Scientist – Intern | Jan 2022 - June 2022

Project: Generate Pre-Curated Question Bank (PCQB) by Question and Answer extraction from articles

- Developed a Transformers-based two-step model for Question Generation followed by the answer extraction
- Scrapped Texts, People Also Ask (PAA) questions and answers using queries related to the E- Commerce domain
- Achieved a Perplexity score of 82.3 on Question Generation by fine-tuning a T5 model on the PAA dataset
- Attained F-1 score of 0.79 on the answer extraction task by fine-tuning the T5-large model on the PAA dataset
- Deployed the two-step model pipeline on the streamlit-based demo web application that accepts user input

Tools and Software: Python, PyTorch, Transfer Learning, PAA, T5 Model, BERT, streamlit

ZS Associates| Data Science Associate – Intern | Jan 2021 – June 2021

Project 1: Extract biomedical text dataset, identify entities, and classify if there exists a relation between entities

- Created a pipeline to extract texts from PubMed database, identifying entities using Selenium and PubTator
- Implemented Binary Classification rules, devised four labeling functions using bio-verbs, co- occurrence of entities
- Generated a training dataset utilizing the four labeling functions in Snorkel by applying the Weak Supervision
- Achieved F1 score of 0.88 on the test dataset in relation-classification by fine-tuning RoBERTa base model

Project 2: Identify the type of relationship between two entities if it exists from the result of the Project-1

- Created a new set of three labeling functions for relation-type identification by using the results of the project-1
- Attained F1 score of 0.83 on the test dataset using XGBoost Model followed by feature engineering

Tools and Software: Python, PyTorch, Transfer Learning, Medline-Plus API, PubTator, Selenium, Snorkel



Research Experience

Emory University | Volunteer Researcher, Atlanta, GA, USA (Remote) | Jul 2022 – Aug 2023

Project: Predict the type of Venous thromboembolism (VTE), from the medical diagnosis and clinical Impressions

- Reduced manual adjudication of dataset by 20 times using pegasus paraphrasing model on sample dataset
- Achieved F1 score of 0.97 in predicting the type of VTE on test dataset by fine-tuning a Bio- BERT model
- Improved F1 score on test dataset by 20 percent by deploying paraphrasing and Bio-BERT finetuning pipeline

Tools and Software: Python, PyTorch, Transfer Learning, pegasus model, BERT

Osaka University | Research Assistant, Ibaraki, Osaka, Japan (Remote) | Jan 2020 - Dec 2020

Project: Predict Non-Small Cell Lung Cancer (NSCLC) using Machine Learning, identify potential drug targets

- Extracted 412 essential genes out of 10,077 by applying Boruta Feature selection on gene expression dataset
- Obtained F-1 score of 1.0 on validation, 0.98 on test dataset by using the XGBoost model to predict NSCLC
- Predicted drug targets for the NSCLC by simulating a Bayesian Network Model on Rho-GDI signaling pathway
- Discovered methodology leads to an accurate treatment of the disease impacting 85% of the lung cancer

Tools and Software: Python, Pandas, NumPy, matplotlib, XGBoost, Bayesian Network, networkX



Competitions and Conferences

- DevCon, Annual Developers and Inventors conference – American Express – May 2025
- The Harvard Project for Asian and International relations (HPAIR) – Hong Kong (SAR) - Aug 2023
- Annual Amazon Machine Learning Conference (AMLC) – Bengaluru, Karnataka - Aug 2022
- 23rd World Business Dialogue, Creation Lab at Evonik - Cologne, Germany - Jun 2022
- Amazon ML Summer School 2021: Offered PPI - Jul 2021
- International Conference on Human Interaction Emerging Technologies - Aug 2020
- Young Data Scientists annual meetup at Kaggle - days, Dubai World Trade Centre - Mar 2020
- Winner at Databuzz 2020 conducted by DoMS, IIT Madras - Jan 2020